

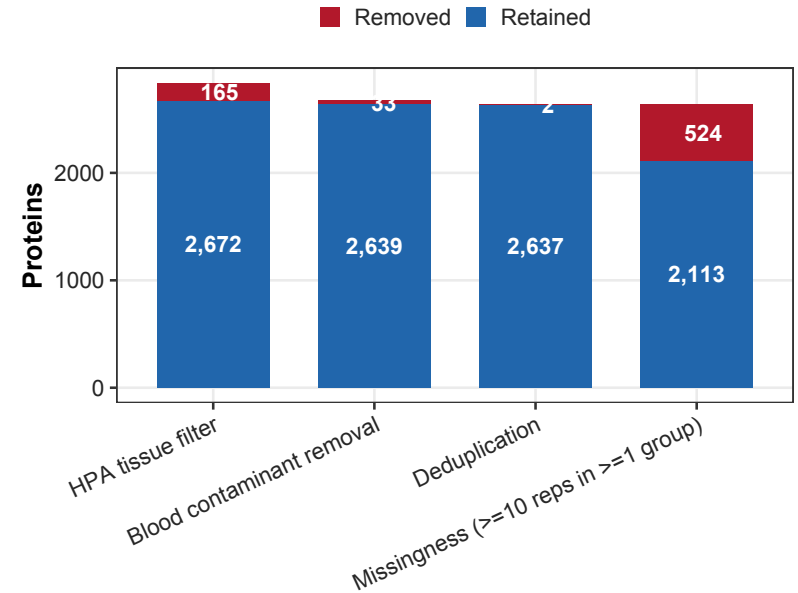
YvO Proteomics Pipeline — Quality Control Summary

00_input -> 01_normalization (cyclic loess) -> 02_Imputation (missForest) | 2,113 proteins x 62 samples | 2026-04-16

A

Protein filter cascade

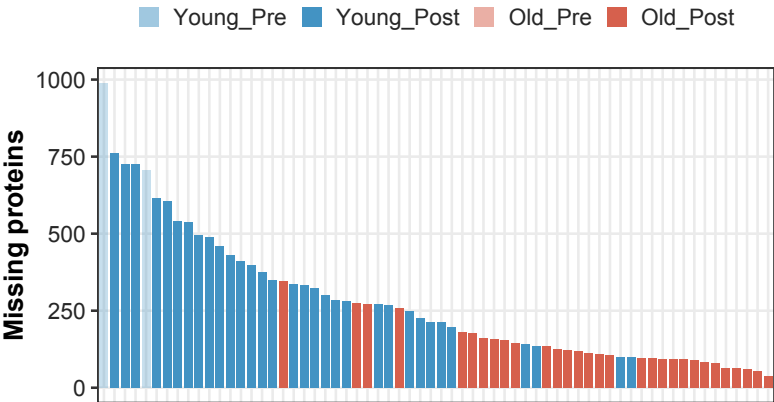
2,837 raw -> 2,113 retained



B

Per-sample missingness

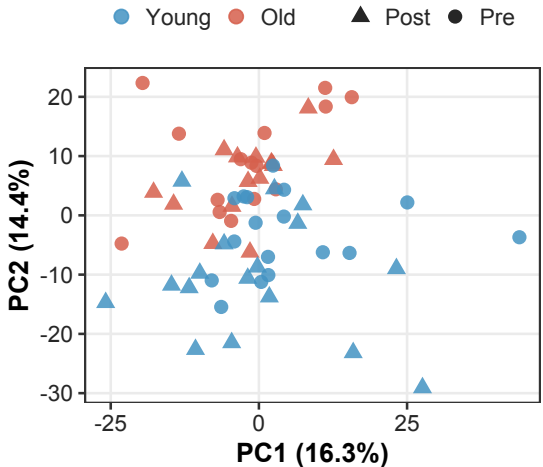
64 samples | consensus outliers faded



C

Pre-normalization PCA

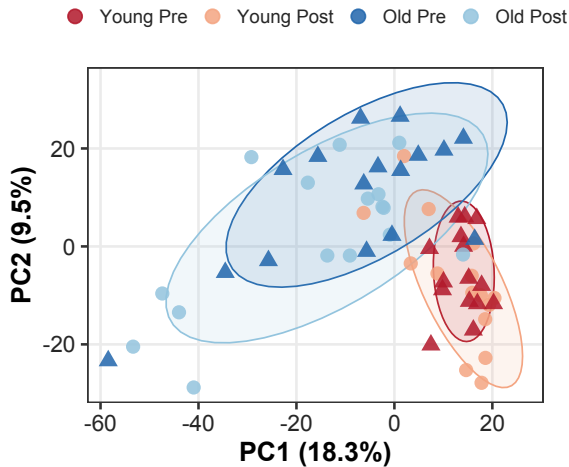
PC1 16.3% | PC2 14.4%



D

Post-normalization PCA

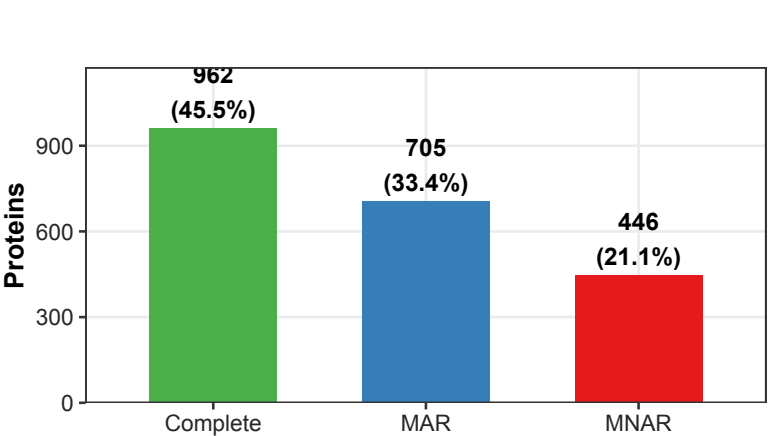
PC1 18.3% | PC2 9.5% | cyclic loess



E

Missingness classification

2,113 proteins | k-means (cluster means: 17.1 vs 19.6) classification



F

Imputation effect

115,173 observed | 15,833 imputed | OOB = 0.109

